

Department of Mathematical Sciences

Faculty of Applied Sciences

Wayamba University of Sri Lanka

B.Sc. (General / Joint Major / Honours) Degree Program

Academic Year 2023/ 2024 – Semester II

MMOD 3214 – Numerical Methods
Tutorial 10

01) Given the following values of $f(x)$,

x	0.3	0.8	1.3	1.8	2.3
$f(x)$	1.0000	1.6487	2.7183	4.4817	7.3891

Approximate $\int_{0.3}^{2.3} f(x) dx$ using the trapezoidal rule.

02) Approximate

$$\int_0^3 (3x^4 + 2x^2 + 1) dx,$$

Using Simpson's rule with $n = 4$.

03) Evaluate $\int_0^\pi \sin(x) dx$ using,

- (i) Composite trapezoidal rule and composite Simpson's rule by dividing the range into ten equal parts.
- (ii) Verify your answer with exact value.

04) Find the approximate value of;

$$\int_0^1 \frac{x}{1+x} dx$$

using (i) Composite trapezoidal rule,

(ii) Composite Simpson's rule,

with six equally spaced subintervals.

05) The exact value of

$$\int_0^1 \ln(1+x) dx$$

is known to be $2\ln 2 - 1$.

Using $n = 6$;

- (i) Compute the approximation using composite trapezoidal rule.
- (ii) Compute using composite Simpson's rule.
- (iii) Calculate absolute error for both methods.
- (iv) Determine the convergence behaviour.

Submit on or before 3rd May 2026 at 12.30 p.m.